

SketchForce AI: Drag-to-Detain – Instant Forensic Suspect Identification from Eyewitness Memory

Vadlamudi Sreevanth Chowdhary
Department of Computer Science and Engineering
Amrita School of Computing
Amrita Vishwa Vidyapeetham, Bengaluru, India
bl.en.u4cse22176@bl.students.amrita.edu

Yendamuri Nishanth
Department of Computer Science and Engineering
Amrita School of Computing
Amrita Vishwa Vidyapeetham, Bengaluru, India
bl.en.u4cse22164@bl.students.amrita.edu

Nandu C Nair
Department of Computer Science and Engineering
Amrita School of Computing
Amrita Vishwa Vidyapeetham, Bengaluru, India
c_nandu@blr.amrita.edu

Tammalla Nikhil
Department of Computer Science and Engineering
Amrita School of Computing
Amrita Vishwa Vidyapeetham, Bengaluru, India
bl.en.u4cse22156@bl.students.amrita.edu

Shaik Adnan Tousef
Department of Computer Science and Engineering
Amrita School of Computing
Amrita Vishwa Vidyapeetham, Bengaluru, India
bl.en.u4cse22053@bl.students.amrita.edu

Abstract—SketchForce AI: Drag to Detain is a computer forensics identification program for law enforcement agencies to produce and identify suspect criminal faces based on eyewitness accounts. Traditional methods of forensic sketching are labour-intensive and often rely on skilled artisans resulting in inconsistencies from artist to artist as well as difficulty in converting traditional sketches into a digital recognition system. The proposed system will solve these problems by providing stand-alone software applications built on JavaFX that allow users to generate standardized face sketches by simply dragging and dropping items into place without creating the end result through artistic work. Upon completion, these sketches will go through preprocessing techniques such as converting colour images to black and white, performing contrast enhancement using CLAHE, and removing background noise with OpenCV in order to increase the ability of the sketches to be recognized when being matched back to actual suspects. After processing, the sketches will be securely uploaded to AWS S3 for security and matching to existing criminal records using the AWS tool "Rekognition" which makes suspect comparisons and provides a similarity score via a deep learning model to compare all submitted sketches against the structured criminal database. The system will employ all appropriate security measures to ensure data is protected from unauthorized use including using secure authentication methods such as one-time password (OTP) validation, JSON Web Token (JWT) session management, BCrypt hashing of all passwords, and IAM controlled access to AWS services. By integrating various technologies such as forensic sketch construction, cloud computing, image processing, and AI based facial recognition will greatly decrease the number of days needed to identify suspects down to minutes, while also enhancing capabilities for efficiency, scalability and/or accuracy in crimes investigations.

Index Terms—Forensic Sketch Generation, Facial Composite System, Criminal Identification, Face Recognition, AWS Rekognition, AWS S3, JavaFX, Drag-and-Drop Interface, Image Preprocessing, OpenCV, CLAHE (Contrast Limited Adaptive

Histogram Equalization), Facial Recognition, Deep Learning, Computer Vision, Digital Forensics, Eyewitness-Based Sketching, Suspect Matching, Cloud Computing, JWT Authentication, BCrypt Password Hashing, One-Time Password (OTP), Identity and Access Management (IAM), Law Enforcement Technology.

I. INTRODUCTION

Eyewitness accounts are typically very influential in helping identify potential suspects in the initial stages of criminal investigative efforts. Eyewitnesses recount their memories of how a particular person looked to using skilled forensic sketch artists to develop a hand-drawn image of a potential suspect. This conventional method of creating forensic sketches can be extremely time-consuming, requires artistic ability and skill, and often produces inconsistent results from an individual to another because each artist may interpret what they see and how they create the images differently (e.g., artist skill level). In addition, many of the sketches created manually can be difficult to upload into modern digital databases used by law enforcement to facilitate criminal investigations and/or into automated systems for facial recognition, thus, limiting the ability to quickly identify possible suspects.

In an effort to overcome these limitations and limitations associated with traditional methods of developing forensic sketches, the SketchForce AI: Drag-to-Detain Project is being developed as a computer-implemented application that takes advantage of AI to build and recognize forensic sketches of possible suspects and will prove to be a more modern and efficient way to identify potential suspects. The SketchForce AI system will be developed as a standalone desktop application using JavaFX so law enforcement personnel can develop

standardized digital composite images (forensic sketches) of potential suspects without requiring professional artistic drawing skills, through the use of a drag-and-drop interface. The facial components used to create a digital composite image of a potential suspect will be organized by category (e.g. eyes, nose, mouth, ears, hair, and facial structure), so that a user can quickly assemble an image of a potential suspect based solely upon eyewitness accounts of how an individual appeared.

The composite sketch is created and then put through preprocessing using OpenCV. The sketch is converted to greyscale, contrast enhanced with CLAHE (Contrast Limited Adaptive Histogram Equalization) and noise removed to enhance clarity and meet the requirements of AI based recognition algorithms. The processed sketch is uploaded to AWS S3 cloud storage where it is encrypted and controlled by means of AWS IAM based access mechanisms. Once uploaded, the composite sketch is then sent to AWS Rekognition for analysis which uses the AWS cloud based deep learning facial recognition system and matches the composite against the criminal mugshot database to return the closest possible suspects with a confidence score.

There are also very strong security measures employed by the system such as Authenticating via OTP, using JWT Tokens for Managing Session and BCrypt Password Hashing, and using HTTPS to ensure confidentiality of personal data and limit access to personal forensic data. Also Ability to Upload and Process Existing Hand Drawn Sketches will enable this project to Continue Supporting Earlier Models. The combined components of the proposed system (forensic sketch generation; cloud computing; image processing; and AI-based facial recognition) will shorten the suspect identification process from many hours and/or days to only a few minutes. This project will provide increased operational efficiency for law enforcement investigations; increased accuracy in matching suspects, reduced reliance on forensic artists, and a scalable, secure, and easy-to-use method for contemporary law enforcement agencies (including the military).

As crime rates rise and there is an increased demand for the timely identification of suspects, the limitations of traditional forensic methods have become apparent. In typical criminal investigations, eyewitnesses give verbal accounts of what the suspect looked like; based on these recollections, forensic artists draw the suspect's likeness. Creating sketches from these accounts takes considerable time and the skills of highly-trained professionals, often results in sketches that differ from each other due to both the creativity of the artist creating each drawing and the ability of the eyewitness to accurately recall the subject. In many parts of the world, due to the small number of forensic artists, delays often occur while investigators wait for the artist to develop a sketch; this ultimately affects how successfully officers can identify a suspect.

Another problem with the traditional method of (manually) creating a sketch for a suspect, is that traditional hand-drawn sketches do not lend themselves to the use of modern computerized facial recognition systems. Existing criminal

databases have been created to match photographs of criminals; therefore law enforcement will often be required to perform manual comparisons between sketches produced from eyewitness recalls and large quantities of photographs. This process is time-consuming, labour intensive and subject to human error. Furthermore, current face recognition technology has been designed to match photographs with photographs only; thus very little support is available for the comparison of sketches with photographs.

Recent breakthroughs in AI, cloud computing, computer vision, and deep learning technology can be utilized to upgrade as many parts of the forensic investigation system. Cloud solutions such as AWS Rekognition provide a scalable solution to facial recognition while improving the quality of the sketches made using image processing techniques. However, very little research has been conducted looking at sketch synthesis and recognition independently with no comprehensive solution that will provide law enforcement with full end-to-end deployable forensic systems.

The motivation for this project is to create a forensic platform that includes an easy-to-use interface for building sketches, automated image preprocessing, secure cloud storage, and AI-enabled suspect recognition as a complete integrated system. This project will reduce the dependence upon forensic artists, shorten the investigative time line, increase the accuracy of matches and provide law enforcement with a fast, secure and accurate means to identify criminals. Further, the assurance that sensitive forensic data is protected and that the integrated platform is secure and scalable by employing proper authentication processes and leveraging cloud-based technologies.

The purpose of this proposed system is to create a more efficient computerized system for improving the efficiency of criminal investigations by producing standardized sketches from eyewitnesses, which can be efficiently matched against criminal database in real time to improve the speed of decision making and enhance public safety.

II. LITERATURE SURVEY

Over the last several decades face-recognition systems both have seen rapid growth in terms of scalability and performance. These systems rely heavily on image processing, as well as computer vision (in particular) technologies. The traditional way of using forensic identification and sketch techniques was primarily through the creation of drawings/sketches of suspects, by forensic artists (or other professionals who create such drawings), and these methods can take a long time, and be somewhat subjective in their assessments. This makes them difficult to implement with today's automated systems that use face-recognition technologies.

Currently, the combination of advances in Deep learning, Cloud computing, and computer vision via tools like Neural Networks (e.g., Convolutional Neural Networks & Generative Adversarial Networks), Arc Photometric Systems/Siamese Networks, as well as cloud-based recognition services (e.g., Amazon Web Services), have provided many new tools for

developing improved methods for generating sketches, matching sketch-to-image (photo), reconstructing a person from their image data, and using Cloud-based technologies to match sketch-to-photo comparisons. Image processing–preprocessing and recognition–using the OpenCV image processing software library and AWS Rekognition cloud service technologies, has contributed greatly to the scalability and performance of modern face analysis systems.

There has been remarkable advancement in facial recognition technologies, but the majority of available solutions either emphasize photographic (photo-based) recognition or isolated sketch synthesis approaches. Only a small fraction of current solutions are able to offer a unified and complete end-to-end forensic framework for creating, preprocessing, cloud storage, secure authentication, and AI-based suspect matching of sketches within one deployable environment. Therefore, it's critical to examine the strengths and weaknesses of existing solutions so that the current research gaps can be identified and an effective forensic sketch recognition system can be developed.

Face recognition and forensic sketching are both significant in terms of popularity and research in the fields of computer vision, artificial intelligence and forensic science because they have been growing in importance to the criminal investigation system and also in public security systems. Traditionally, suspects have been identified using manually created sketches and descriptive characteristics of the suspect by independent witnesses, which can be very tedious, subjective and difficult to integrate into an automated digital recognition system. Many different techniques such as cloud computing, AI-based face recognition, deep learning, image processing and sketch synthesis have been explored by academic researchers in order to address these issues.

Guo et al. [1] developed the domain alignment embedding network to minimize the difference between traditional forensic sketches and true facial photographs. In this framework, the authors proposed the use of a domain alignment loss function that enhanced the consistency of the image features between sketches and photographs when recognizing the subjects. Compared with traditional CNN-based techniques, this framework exhibited improved performance in sketch to photograph matching; however, it requires a high degree of computational resources, large amounts of training data and complex optimisation techniques, which severely limit its ability to be used for real-time recognitions in practical forensic environments. Jambur et al. [2] discusses the reconstruction of facial features using Generative Adversarial Networks (GAN) for image generation and CCTV image enhancement. The proposed GAN system was able to improve the quality of low-resolution surveillance images and produce reconstructed facial images for later evidentiary purposes. The results show that while their approach did enhance the quality of reconstructed facial images, the focus was primarily on visual restoration and not on a fully integrated sketch-to-database matching system for real-time identification of potential subjects.

Bagwani et al. [3] study they analysed the various cloud machine learning APIs like AWS Rekognition. The result of this study showed that cloud platforms could significantly improve the face detection and recognition tasks, as they allow for faster processing times, enhanced scalability and more flexibility in terms of using less local hardware. However, their proposed solution was focused primarily on recognizing people by their direct photographs, since they did not consider how forensic artists create sketches or how to match sketches to photos. Fuad et al. [4] provided a comprehensive review of modern methodologies to perform face recognition using Convolutional Neural Networks (CNNs), Generative Adversarial Networks (GANs), transfer learning, and hybrid architectures of recognition. Feature extraction methods, alignment of faces before recognition, and improvements made to the accuracy of recognized faces were discussed in their survey. Although this article provided some helpful information about modern face recognition methods, it did not specifically address the recognition of forensic drawings, or how law enforcement could implement an operational system for such purpose.

Aruna Bhat et al. [5] developed the research project "Robust Face Detection and Recognition using Image Processing and OpenCV." The study used classical image processing methods implemented through the OpenCV library and Haar Cascade classifiers to detect and recognize human faces. The research resulted in successfully identifying human faces under closed-controlled conditions and provided effective performance when processing standard images of human faces. However, the study could not perform reliably with variable light conditions, orientations of the human face, and different qualities of photographs utilized during the analysis of human faces for recognition purposes. Also, the research study did not support the use of forensic sketches, nor did the study support the use of advanced deep learning techniques. Ankit et al. [6] published "Sustainable and Secure Real-Time Face Recognition and Notification System using AWS Serverless Architecture." This research project designed a system that integrated AWS services like AWS Lambda, AWS Rekognition, AWS S3, and SNS to create a real-time facial recognition and automated notification system. The research demonstrated scalability, reduced infrastructure management needs, and the ability to communicate via the cloud. However, the system was designed solely for direct photo-based recognition and did not support forensic sketches or the comparison of sketches to photographs.

Mohanad et al. [7] Proposed Face Detection Classification Model Based on Amazon Web Services Cloud (AWS). The authors utilised CNN and MTCNN architecture with AWS Cloud infrastructure to enhance accuracy in detecting faces through computational efficiency associated with AWS Cloud technology among other advantages previously mentioned. The system exhibited increased accuracy of recognition and lower latency regarding processing time. Unfortunately, the authors only focused on face recognition within their work and did not support the generation of forensic sketches or match to criminals from database records. Shivani et al. [8] wrote an

article evaluating "Enhanced Reconstruction of Digital Images and Contextual Intelligence for Cyber-Forensics". Their article presented a method to enhance digital images through the application of super-resolution algorithms in conjunction with contextual intelligence to provide better forensics evidence. The authors described improvements in clarity and overall reconstruction of degraded digital forensics images through the use of their methods; however, their work did not examine or identify facial sketch matching based on a witness' description.

Ling et al. [9] conducted a study titled 'Exploration of Computer Vision and Image Processing Technology Based on OpenCV'. The authors utilized traditional approaches to implement an OpenCV-based facial detection and image processing pipeline designed for the purpose of performing recognition tasks. The results yielded a highly efficient method of edge detection, facial extraction, and preprocessing. The limitation of this study is that it utilized graphical processing units (GPU) primarily for classical methods of image processing, and thus, were not able to utilize advanced methods of image processing based on deep learning using neural networks to perform sketch recognition. Poornima et al. [10] published a study titled 'Face Sketch Recognition Using Image Synthesis and Deep Learning'. As part of this research, they combined the use of style-based GAN (generative adversarial network) sketch production with the use of CNN (convolutional neural network) image detection. The authors demonstrated that using sketch-based production and facial identification together improved face photo recognition for the generated sketch images. One limitation associated with using this architecture includes a lack of generalization capability in regards to real-world forensic sketches, in addition to the absence of cloud scalability and secure deployment options.

Xingqun et al. [11] proposed Face Sketch Synthesis via Semantic-Driven Generative Adversarial Network. The method used a generative adversarial network (GAN) architecture where a facial identity of a person was preserved during sketch generation by embedding semantic priors into the sketch synthesis process. The resulting sketches were semantically consistent with the input sketches and semantically consistent with sketches generating a matching identity. The work was concerned with sketch synthesis only and could not perform recognition or matched with a criminal database on a real-time basis. Fan et al. [12] proposed An Improved Siamese Network for Face Sketch Recognition. This work focused on using a siamese CNN architecture to help bridge the modality differences between sketches and photographs through learning of representative feature embedding. Their results showed the improved performance of the matching of sketches in benchmark datasets over previous techniques. However, although both datasets created by using this method had been carefully balanced; both methods were limited in their usability for real-time forensics and scalability to cloud-based methods.

Kamruzzaman et al. [13] developed an "Automated Attendance System for the Whole Classroom with Multi-Face Facial Recognition using AWS Rekognition." Their research

shows the potential for efficient multi-face recognition using AWS Rekognition to enable the creation of an automated attendance management system. The cloud-based architecture of the proposed attendance management system would allow for scalability and automation but does not offer the ability to input a forensic sketch or use the system for criminal investigations. Sritama et al. [14] performed a study titled "Cloud-Based Facial Recognition System Architecture using AWS Services: A Complete Analysis." The purpose of this study was to conduct a comprehensive analysis of scalable cloud-based facial recognition architectures using AWS services. The researchers concluded that there are significant advantages associated with the use of cloud computing as it relates to the storage capacity of the system as well as its ability to scale out and have distributed processing occur. The proposed architecture examined by the researchers uses standard facial images for recognition and therefore does not have the capability to recognize a forensic sketch.

Lei et al. [15] developed a cross-modal learning architecture that aligned the features of sketches and photographs (facial images) through identity-discriminative learning. The model used this cross-modal learning process as a means to "bridge the gap" between the two different modalities of sketches versus photographs as well as to improve the accuracy of recognition-based on direct photograph to sketch recognition. Although the architecture was successful at bridging the gap and improving the overall accuracy between a sketch and a photograph, the architecture was very complex to compute while lacking practical interfaces for the actual creation of a sketch or the use of a cloud platform for the deployment of the created sketch. Alen et al. [16] developed an Android application called "Development of Android application for gender, age and face recognition using OpenCV." This Android application utilized OpenCV based recognition algorithms to detect users' facial attributes, including age and gender, and provided mobile recognition capabilities. While this application performed well for recognition of facial features from a direct photograph, it is limited to using photograph to only recognize facial attributes and therefore does not have the ability to recognize sketches for forensic evidence.

Weiguo et al. [17] introduced a framework called "Generative Adversarial Multi-Task Learning for Face Sketch Synthesis and Recognition." This new architecture is a multi-task GAN (generative adversarial network) developed to facilitate both the synthesis of sketches as well as the recognition of sketches concurrently. The proposed framework narrows the gap between the modalities of sketch and photo image processing, as well as improving the performance of recognition for both sketch-to-photo and photo-to-sketch. The disadvantage to this framework is that it will require a large amount of resources in order to train the model, a very large dataset, and no ability to deploy securely in a cloud-based deployment, as well as no ability to generate sketches interactively.

Overall the major conclusion from the review of the literature is that there is a significant amount of research on each of these isolated tasks, such as sketch creation, photo-based

recognition, cloud-based detection, image enhancement; but there is limited, if any work that has been done in creating an integrated forensic capability, which includes drag-and-drop sketch generation, pre-processing, secure authentication, cloud storage, and AI-based suspect identification, into a single operating environment. The significant deficiencies in the existing research emphasize the necessity for a secure, scalable, and practical forensic sketch recognition system, such as has been proposed in this project.

It is apparent from the literature review that almost all of the currently available face- recognition systems and those involved in the analysis of forensic sketches have significant limitations that prevent them from being used as tools to assist law enforcement in investigating real crimes. There has been a great deal of research conducted into developing deep-learning models for sketch recognition and cloud- based recognition models; however, there are only a few options that provide a practical, comprehensive, end-to-end solution for law enforcement agencies to use in the field.

A significant research gap that has been identified is the lack of a holistic integrated forensic sketch generation system. Almost all current methods for generating forensic sketches focus solely on sketch synthesis or face recognition with no integrated system that provides features such as drag-and-drop sketch creation, preprocessing, cloud upload, and suspect identification all within a single environment. In addition, most forensic sketch-generating systems still rely heavily on the expertise of forensic artists, leading to a lengthy and difficult investigation process overall.

The large visual difference between a forensic sketch and an actual photo of a person's face is another major limitation in this area. Most current face recognition models work well on quality-controlled data sets but have extreme difficulty working with low- quality forensic sketches. The models also struggle with varying head positions, inconsistent representation of facial features, and real-life conditions where most forensic sketches need to be identified. Although many deep learning architectures (GANs / Siamese Networks / CNNs, etc.) can improve face recognition performance, they often require extensive resources to train and run, making them impractical for widespread use.

Cloud-based services for face recognition like AWS Rekognition and similar APIs are designed to recognize faces primarily in photos and provide little, if any, direct support for matching facial sketches. There is very little existing work that provides any form of integration between cloud-based infrastructure and processing pipelines for forensic sketches so there is still a lack of connection between sophisticated face recognition algorithms and scalable real-world deployment of such systems.

Now, there are also concerns about security and usability concerning the current forensic systems being used today. Many of today's current forensic systems lack adequate secure authentication methods, encrypted communications and controlled access to the cloud to contain sensitive data for a criminal investigation. In addition, many research-based

forensic systems do not have a user-friendly interface for investigators who may not have any technical experience to easily operate those systems.

Another gap in forensic systems today is a lack of backward compatibility to work with traditional forensic methods, for example: the majority of today's modern forensic recognition hardware systems cannot process any of the hand-drawn forensic sketches located in the police records; therefore, they are not compatible with the current criminal databases and investigation workflows.

The AI-based forensic sketch recognition system intends to resolve these gaps by integrating drag and drop sketching capability, use of OpenCV to process photographs and use of secure cloud storage, One Time Password (OTP) verification for authentication to access the forensic database, recognition of suspects using AWS Rekognition, and scalable cloud infrastructure to establish a single forensic investigation platform. This forensic sketch recognition system will provide modern law enforcement with a secure, practical and real-time sketch-to-database match capability.

III. SYSTEM DESIGN

A. Framework of the Proposed System

The AI Based Forensic Sketch Recognition System is a secure intelligent forensics platform that provides an integrated workflow to facilitate the sketch drawing, preprocessing image, storage (cloud), AI-based recognition of the suspects through single window access and thus allows for law enforcement agencies to expeditiously determine the identity of a criminal suspect based on eyewitness account with reduced manual intervention and limited dependence of a forensic artist.

The system is based on multiple interconnected modules for User Authentication, Sketch Building, Image Enhancement, Storage (Cloud), Feature Extraction, Recognition, and Output of Results. The Workflow process starts with the law enforcement officer logging in through a secure method utilizing a user name/password combination and a OTP-based 2FA process. After successful log in, a user can create a suspect sketch by dragging and dropping the facial characteristics or uploading a pre- existing rough sketch into the system.

The sketch constructed by the officer is processed using Image Enhancement techniques to improve the likelihood of identifying the suspect. The enhanced image is uploaded securely into AWS Cloud and compared against the suspect images using AWS Rekognition AI-based matching. The system will retrieve and display the authentic Matching Criminal records and the similarity confidence scores and any related case information.

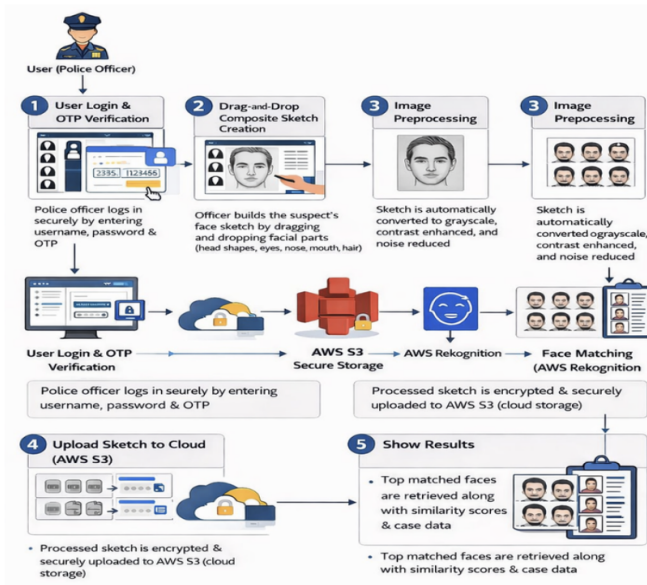


Fig. 1. Framework of the AI-Based Forensic Sketch Recognition System

The proposed system (see Fig. 1) is demonstrated by the full operational workflow. This process begins with the secure log in of a user, requiring two-factor authentication (2FA) in order to ensure only authorized users may have access to the forensic application. Once the user has been authenticated, the police officer creates a composite suspect sketch through a drag and drop approach utilizing various facial components (i.e., head shape, eyes, nose, mouth and hair). After the composite suspect sketch has been created, the system performs image preprocessing functions (e.g., conversion to grayscale, contrast enhancement using CLAHE, and noise removal) that improve the clarity of the sketch and make it conducive to face recognition algorithms.

Once the identified composite suspect sketch is pre-processed, it is then securely uploaded to AWS S3 (Cloud storage) with cloud-based security (encryption) in place that will protect the sensitive forensic data. AWS Rekognition then analyzes the suspect sketch which was uploaded, and utilizes deep learning based facial recognition techniques to compare the suspect's sketch against known criminal mugshots stored in database(s). If a match is found, the reconnaissance module retrieves the closest matching suspects and provides all case details associated with each suspect with the applicable similarity percentages. This operational workflow significantly reduces the time frame in which suspects may be identified during an investigation and adds to the overall efficiency of investigations.

B. Sketch Construction Module Design

The proposed system will include a sketch construction module that serves as one of its core components. The previous methods of forensic sketching require a great deal of time and skill from artists in order to generate an accurate representation of the suspect. To remedy this disadvantage, the proposed system will incorporate a drag- and-drop facial composite

generation tool that will allow law enforcement personnel to use standardised sketches without any drawing ability.

The sketch construction module allows for users to select from a catalogue of facial categories, such as head shape, eye shape, nose shape, ear shape, lip shape and hair type. All of the categories will be predefined and stored within a database that is part of the proposed system. Any of the feature selections can then be modified, either by resizing, moving or replacing them based upon eyewitness accounts. The sketch creation process will be an iterative one, allowing investigators to continually refine their image of what the suspect will look like until the final sketch has been created.

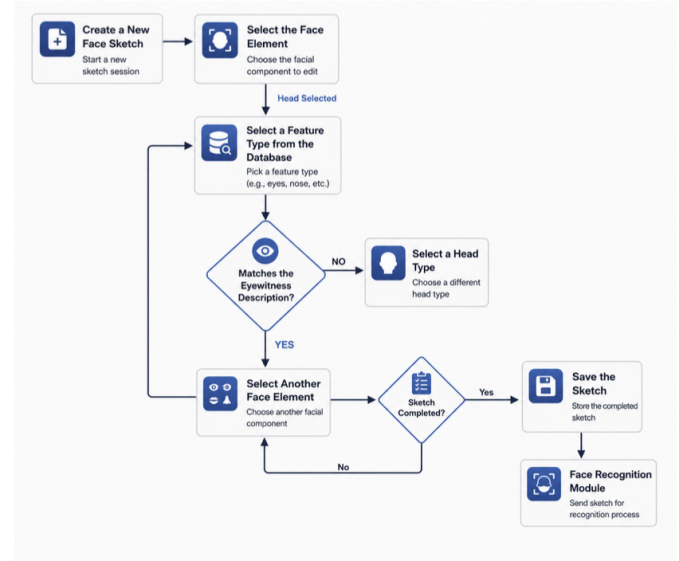


Fig. 2. Workflow of the Composite Sketch Construction Module

Sketch construction follows a workflow shown in Fig. 2. The first step is to initiate a new sketch session. Once the new sketch session is initiated, the user selects a feature from the repository of available features. The system will then match the feature selected by the user with the description of the suspect provided by the eyewitness and whether or not the feature matches the description; If it does not match, another feature from the feature repository can be selected. After a user selects and approves a feature, additional features may be selected and combined to create the final product of the completed suspect sketch.

The entire workflow process is repeated until the total number of necessary facial elements has been assembled into the completed suspect sketch. Once the final sketch has been generated, it will be stored and sent to the face recognition module for analysis. The goal of using this structured workflow is to provide uniformity in the creation of the suspect sketch and limit the reliance on the highly trained forensic artist for sketch generation.

C. Recognition and Identification Module Design

The recognition and identification module is responsible for processing the generated sketch and matching it against

Criminal databases using cloud-based AI services. This module integrates image preprocessing, feature extraction, cloud-based storage, and facial recognition into a secure and scalable forensic pipeline.

The generated sketch is first processed through image preprocessing operations using OpenCV techniques. The generated sketch will be converted to a grayscale file type to remove any unnecessary color information and to reduce the computing complexity of the processing. To enhance the visibility of the facial structures of the sketch, such as nose, eyes, lips and jawline, Contrast Limited Adaptive Histogram Equalization (CLAHE) will provide a contrast enhancement to the generated sketch. Furthermore, the use of noise-reduction techniques will allow the removal of any distortion from the generated sketch, thus smoothing out the image. These preprocessing operations increase the recognition accuracy of the generated sketches by increasing the capabilities of the forensic generated sketch to be recognized by digital face recognition systems.

After all preprocessing operations have been completed, the generated sketch will be securely uploaded to Amazon Web Services (AWS) Simple Storage Service (S3) cloud storage. AWS Identity and Access Management (IAM) policies along with Hypertext Transfer Protocol Secure (HTTPS) communication protocols will provide secure communications for the transmission of the sketch data, and controlled access to the forensic data. After the sketch has been stored in the AWS S3 cloud storage, AWS Rekognition will extract the facial embedding from the generated sketch and compare them to the mugshot images stored in the criminal database.

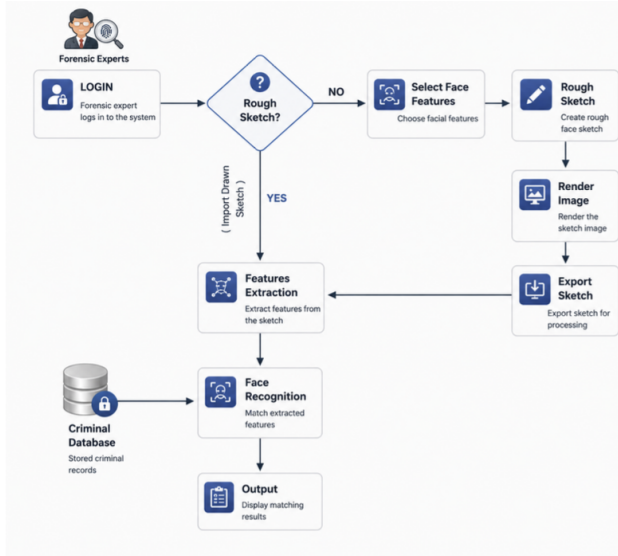


Fig. 3. System Flow for Recognition and Criminal Identification

The workflow for the recognition and identification of suspects is shown as a diagram in Fig. 3. First, the forensic personnel will log into the application as the beginning point of the workflow. Next, the system will determine if there is already a rough sketch available for the user. If there is

an existing hand-drawn sketch available, that sketch will be input directly from the feature extraction module. If not, the users will create a new sketch by selecting facial features and generating a final image.

After the feature extraction process occurs, the recognition module will compare the extracted facial features to the existing criminal records that are stored in the database. Then, using AWS Rekognition, a deep learning-based match is run between the extracted facial features and the existing criminal database to find the closest matching suspect. The final outcome of this recognition module will display the recognition results for law enforcement to analyze the retrieved matches to determine the next steps for investigation.

This workflow also allows for backward compatibility by allowing users to upload and process traditional hand-drawn sketches in the same pipeline as the AI-based suspect recognition and identification system. This allows for a seamless blending of both major classification systems: traditional forensic techniques and modern methods of suspect identification through artificial intelligence (AI).

D. Design Characteristics of the Proposed System

Compared to traditional forensic sketching or methods of identifying suspects manually, the architecture of the proposed forensic sketching system is theoretically superior. It has an ability to generate digital sketches of suspects rapidly via drag-and-drop capabilities creating a digital copy of the original sketch as opposed to relying on forensic artists for creating sketches; therefore investigators can quickly create standardized digital sketches. In addition, it is designed to work seamlessly with Amazon Web Services cloud services which provide secure storage and scalable computing resources, and have capabilities for artificial intelligence (AI) & provide methods that enable investigators to recognize individuals through image comparison quickly and efficiently. The preprocessing function of the system will improve the quality of the sketches generated from the original photographs and improve the probability that each individual will be matched with their suspect sketch. Finally, there are methods available for authenticating access, including OTP verification, JWT authentication, and encryption of communications to limit unauthorized access to these sensitive forensic data.

The system design is modular in nature and enhances the ability to maintain, scale, and operate more efficiently. Therefore by integrating AI, image processing, cloud computing, and forensic investigative techniques into a single platform, the proposed forensic sketching system will increase the efficiency of identifying a suspect and will expedite the process of conducting criminal investigations.

IV. SYSTEM IMPLEMENTATION

The actual implementation of the AI-Based Forensic Sketch Recognition System emphasises combining elements of secure authentication/authorization, forensic sketch generation, cloud-based storage of sketches, image pre-processing, and AI-based recognition of suspects into one desktop application. The

system was implemented using JavaFX for the GUI, Core Java for back-end processing, OpenCV for image processing, SQLite for local database management, AWS cloud services (for storage) and the use of facial recognition functionality. Implementation of the system was divided into several functional modules where each module carries out a particular operation in a forensic investigation workflow. Together these functional modules provide secure access control to a sketch, an effective and efficient method for generating forensic sketches, a secure method for communicating with the cloud, and an accurate method for identifying suspects through the use of artificial intelligence techniques.

A. Authentication and Secure Login Module

The authentication module gives safe access to the forensics system. Because the forensics system has very sensitive information about criminal investigations, there are multiple layers of security to keep people from getting into the system who should not be there (not authorized). The forensics system requires users to login in order to authenticate, as well as to provide an OTP to gain access to the application.

The authentication module also collects information about the device accessing the forensics system, such as the MAC and IP address, to improve the authentication and monitoring of the device. If someone has obtained a user's authentication information but is not the user, their ability to access the system can be limited through the use of OTP verification (in this case, through SMS text message).

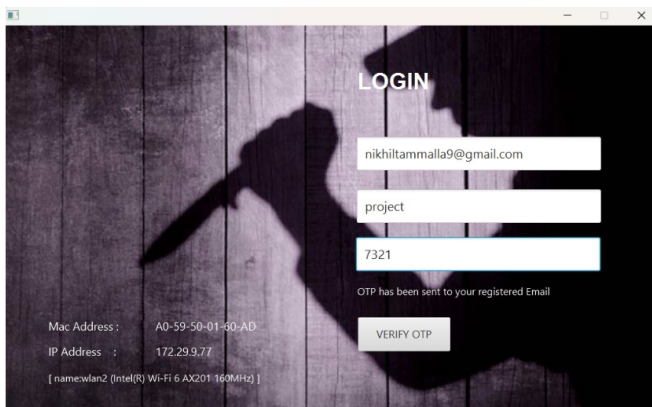


Fig. 4. Login and OTP Verification Interface

Fig. 4 illustrates the secure login interface of the proposed system. The user enters the registered email address and password, after which the system sends a One-Time Password (OTP) to the registered email account. The OTP must be entered correctly for successful authentication. The interface also displays system identification details such as MAC address and IP address, providing additional security monitoring capabilities for the application.

B. Database and User Credential Management Module

The interface for securely logging into the system as described in the attached image (Fig. 4) requires the entering of

both a username and password, then the sending of the OTP (One Time Password) to the provided username/email address before authentication will be considered successful once you properly enter your OTP. It also contains additional monitoring capabilities through identification capabilities via displaying the MAC and IP address for additional information related to each system and thus allowing for additional monitoring capabilities for the application.

The database management module is comprised of all the information necessary to securely log into the application, including login credentials (username and password), information related to the system, and history of all records related to authentication in addition to the real-time use of the application. SQLite is the lightweight database system used to store user login data as well as for keeping records that apply to application-level operations.

The module contains information such as registered user's email account, encrypted password, MAC address, and IP configuration of established users, all of which will be utilized to authenticate a user via OTP and to verify a device's integrity before establishing a user's connection.

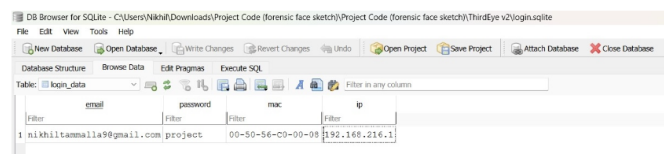


Fig. 5. Database Structure for User Authentication

Fig. 5 displays the SQLite DB table used to store user authentication data, such as email address, password, MAC address, and IP address. The database is designed to allow secure credential verification, as well as to enforce controlled access to the forensic platform.

C. Option Selection Module

Once the user has been authenticated, they can continue to the choice selection module, which allows them to either create a new composite sketch or upload an already created rough sketch into the system. The module also supports legacy functionality (i.e., using traditional hand-drawn sketches) to process these sketches through the artificial intelligence-based recognition pipeline.

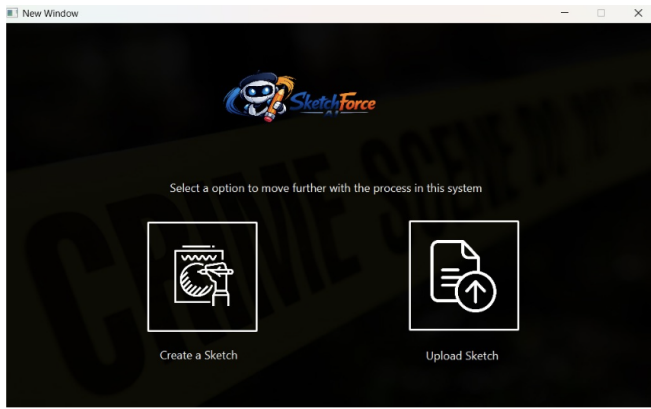


Fig. 6. Main Option Selection Interface

Fig. 6 depicts the interfaces of the choice selection module. There are two primary functions of this module: to create a sketch or to upload a sketch. Selecting the first function will navigate the user to the drag and drop sketch creation module, and selecting the second will allow the user to upload previously created hand-drawn sketches for processing and analysis.

D. Composite Sketch Construction Module

The module that allows an investigator to create a forensic facial sketch using drag and drop facial characteristics, rather than having to hire a professional forensic artist, is a core component of the overall system implementation described in this document.

The different facial characteristics are organized in different categories such as head shape, hair, eyes, eyebrows, nose, and mouth. Users can create a forensic facial sketch by using eyewitness descriptions to find and choose facial characteristics from each category, then drag those characteristics into place on the canvas area of the interface, according to the witnesses' statements, until all of the characteristics necessary to create the final facial sketch have been placed on the canvas.

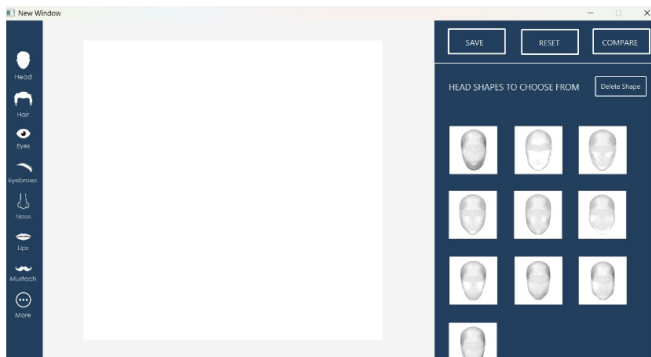


Fig. 7. Composite Sketch Construction Interface

Fig. 7 illustrates the drag and drop interface for creating the forensic facial sketch described in this section. The drag and drop categories of facial characteristics are displayed in the

left pane of the user interface. The actual characteristics will be displayed in the right pane of the user interface, and users select a characteristic they wish to drag and drop on the canvas area of the user interface. Options for saving, resetting, and comparing facial sketches are also available while creating a forensic facial sketch using an eyewitness' description.

E. Dashboard for Face Recognition and Sketch Selection

Once a composite sketch is generated, the investigator goes to the face recognition dashboard (i.e., the module that sits between the sketch generation and the AI-based identification of a suspect by the system). The investigator can select a sketch image, upload it to the cloud server, and begin the recognition process on the face recognition dashboard.

Face recognition provides investigators with three different options on the system's dashboard that enable them to do so; they can select, upload, and identify a suspect. The investigator will be able to perform these actions in an orderly fashion.

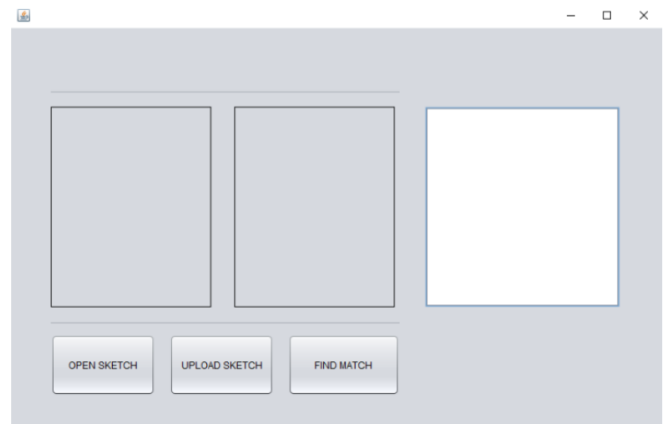


Fig. 8. Dashboard to Recognize Face in Database

Fig. 8 shows the actual interface of the face recognition system used to find suspects in the criminal database. Within the dashboard interface are three areas showing the picture of the uploaded sketch to the identification system; the matched image of the suspect's face; and the results of the identification process. When an investigator first enters the dashboard to identify a suspect, the interface will contain no information until he/she selects and uploads a forensic sketch image for identification.

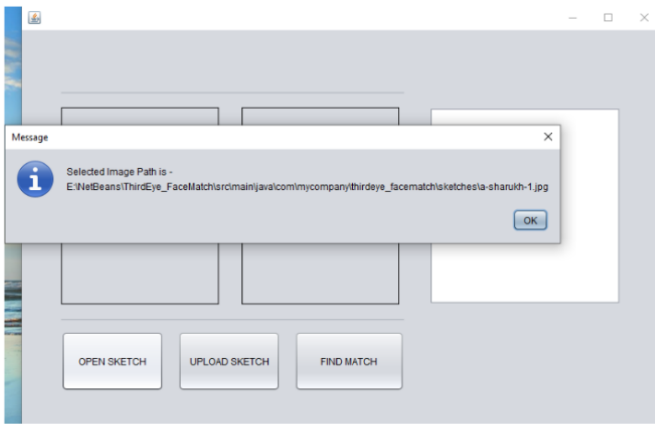


Fig. 9. Select and Open a Face Sketch

Fig. 9 shows the system steps involved in selecting and opening a forensic sketch within the face recognition dashboard. The investigator clicks on the “OPEN SKETCH” button to select the sketch image, at which time the application will show the path to the selected image. The investigator must complete this step before the sketch is uploaded to the cloud for matching, to ensure that the correct sketch is loaded into the identification system.

F. Face Recognition and Matching Module

The purpose of the facial recognition & matching module is to identify a suspect by comparing a generated forensic sketch and a criminal record in the database. The module includes image pre-processing techniques, an ability to securely upload images to the Cloud, and the use of the AWS Rekognition facial analysis service to perform sketch-to-photo matching in a very efficient manner.

When a generated or imported sketch is provided, it first goes through a pre-processing operation via OpenCV. This includes processes such as conversion of the sketch to grayscale; enhancing contrast in the sketch using CLAHE; and eliminating noise from the sketch. Pre-processing enhances the quality of the sketch image, therefore augmenting the ability of AI-based facial recognition algorithms to function with the sketch.

After pre-processing is completed, the pre-processed sketch image is securely uploaded to the AWS cloud using AWS S3 services. After the completion of the upload to AWS, facial recognition is performed using AWS Rekognition. The AWS Rekognition engine takes a sketch and generates facial embeddings of that sketch, compares them to each/any of the photographed criminals stored in the database; and returns the closest matching suspect hits and their associated similarity percentage, confidence score, and other record information.

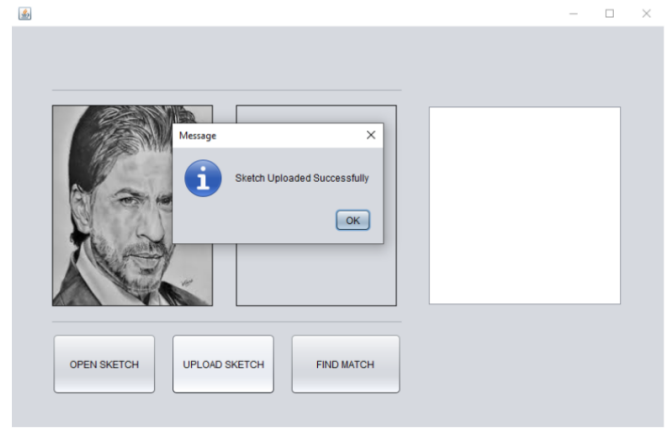


Fig. 10. Sketch Upload to Cloud for Recognition

The initial step in the Recognition component is the upload-ing of a Forensic Sketch to the AWS (Amazon Web Services) Cloud Platform. The Forensic Sketch will first need to be up-loaded using the “UPLOAD SKETCH” button that is included in this component. After the sketch is uploaded successfully, a message will appear on the screen confirming that the sketch has been uploaded successfully to the AWS Cloud via a secure connection for further recognition processing.

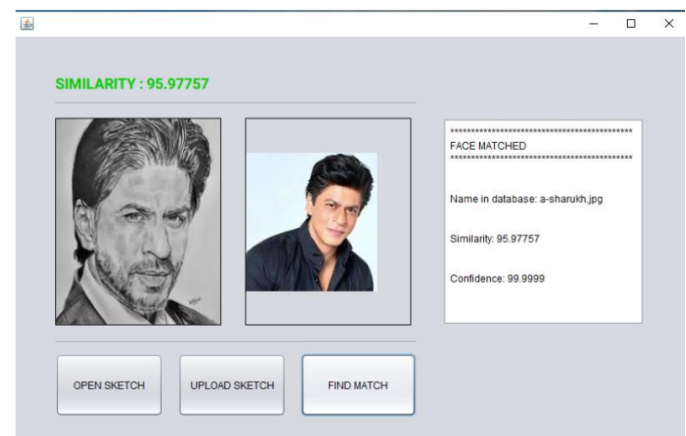


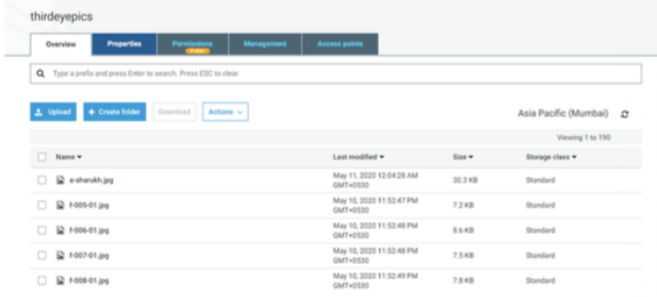
Fig. 11. Face Sketch Matching and Recognition Result

The final Recognition Result generated by the Recognition system after completing the matching process is illustrated in Fig. 11. The Recognition Results page consists of the left side, which displays the uploaded sketch forensic. The center of the screen will show the matched photograph contained in the criminal database that correlates with the uploaded forensic sketch. On the right side of the screen, you will see the identifications and percentages associated with the match (record name, similarity percentage, and confidence scores generated by the AWS Rekognition). This verification allows investigators ease of reference to identify possible suspects and will allow the continued investigation of suspects through the criminal justice system.

G. Police Record Management with Face Images and Details

The proposed solution uses AWS Cloud storage to store criminal face records supporting centralized and scalable identification of suspects, through stored criminal mug shots and associated metadata which are securely stored on the server for retrieval upon request during facial recognition operations.

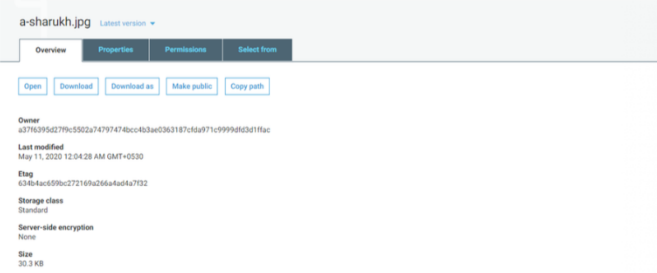
Stored records used by AWS Rekognition compare uploaded forensic sketches to criminal face datasets and return matching suspects based on similarity in the facial images.



Name	Last modified	Size	Storage class
a-sharukh.jpg	May 11, 2020 12:04:28 AM GMT+0530	30.3 KB	Standard
f-005-01.jpg	May 10, 2020 11:52:47 PM GMT+0530	7.2 KB	Standard
f-006-01.jpg	May 10, 2020 11:52:48 PM GMT+0530	6.6 KB	Standard
f-007-01.jpg	May 10, 2020 11:52:48 PM GMT+0530	7.5 KB	Standard
f-008-01.jpg	May 10, 2020 11:52:49 PM GMT+0530	7.6 KB	Standard

Fig. 12. Police Record with Face Images

The criminal face image records stored on the AWS cloud server are shown in the diagram in Fig. 12. There are several facial image datasets stored on this server that support matching operations when identifying suspects. Stored records of criminal records create the primary database for AWS Rekognition-based Face recognition analysis.



Property	Value
Owner	a37b695d279c5502a74797474b0c63a0263187c5d971c999d63d1ffac
Last modified	May 11, 2020 12:04:28 AM GMT+0530
Etag	63b4ac659bc72216a26a4a4a702
Storage class	Standard
Server-side encryption	None
Size	30.3 KB

Fig. 13. Police Record with Face Image Details

Fig. 13 presents detailed metadata associated with a criminal face image stored on the server; metadata includes: file name, location and type of storage, image properties, time of upload, and configuration of storage. Metadata for each of the records creates organized criminal datasets for use in identification of suspects through the use of cloud-based forensics.

V. PERFORMANCE EVALUATION OF THE PROPOSED SYSTEM

A. Evaluation Methodology

The evaluation determined whether the AI-based forensic sketch recognition system has the ability to produce forensic sketches securely; preprocess the images; execute suspect

matching through a cloud environment; and retrieve reliable recognition results. The evaluation involved assessing the efficiency, reliability, scalability and recognition performance of the system as an integrated forensic solution.

The forensic sketch recognition system evaluated 10+ facial sketches created through a drag-and-drop sketch construction module, as well as a number of rough sketches imported into the system. The sketches were preprocessed using OpenCV preprocessing techniques, and then these images were matched against criminal images via AWS Rekognition. The evaluation was based on the following aspects: quality of preprocessing; accuracy of recognition; security of authentication; response time when accessing databases in the cloud; and overall efficiency of the system when executed.

The evaluation environment comprised a JavaFX-based desktop application that is integrated with AWS Cloud services such as AWS S3 and AWS Rekognition. Multiple different configs of facial sketches with different feature combinations and quality were used in order to evaluate the validity and consistency of the recognition procedure.

B. Image Preprocessing Performance Evaluation

Image preprocessing improves the quality of sketches as well as the compatibility of forensic sketches with digital face recognition systems. Since hand-created or composite sketches often have low contrast, background noise, and inconsistent visibility of features, it is necessary to perform some preprocessing work before facial recognition occurs.

The system proposed will use OpenCV integrated into Java to conduct the operations necessary for preprocessing (grayscale conversion, CLAHE-based contrast enhancement, and noise reduction). Converting a sketch to grayscale will eliminate all of the unnecessary color information from the sketch and also reduce the overall computational complexity during feature extraction. Contrasting large areas of a facial structure (eyes, nose, lips, and jawline) will improve local contrast, improving the visibility of these important features. Finally, the techniques used to remove noise will help smooth out the visual distortions that result from unwanted noise while maintaining critical edges of the facial shape.

The preprocessing module greatly improved the visibility of facial features and increased the accuracy of AWS Rekognition in matching sketches with photographs. Testing has shown that enhanced sketches produce higher-scoring similarity scores and improved quality of extracted facial features than have unprocessed images.

C. Cloud Recognition Performance

The system proposed above uses the Amazon Web Services (AWS) Cloud to securely store information and to use artificial intelligence (AI) for facial recognition. The images of criminal sketches and the images for a criminal database were stored in Amazon Simple Storage Service (S3) and facial recognition was provided using Amazon Rekognition.

The sketches of the sketches being processed were uploaded to S3 using an encrypted communications protocol. After

uploading the sketches to S3, Amazon Rekognition would take the uploaded sketches and generate a set of facial embeddings and compare them with the images in the criminal database. Amazon Rekognition also generated a set of similarity scores and confidence values based upon the facial feature matching results.

The cloud-based recognition architecture increased the scalability and processing efficiency of the system. The cloud-based architecture reduced the client/server option of the system by eliminating the need to rely on local computational resources for facial recognition. The speed and accuracy of facial recognition using Amazon Rekognition was very good, producing consistent high confidence matches for sketches which had been accurately preprocessed. The integration of the forensic records into one location with direct access has also improved the centralized storage of forensic records.

D. Authentication and Security Evaluation

Security is an important aspect of forensic investigation systems due to the sensitive nature of criminal data and suspect records. The proposed system incorporates multiple security mechanisms to ensure secure access and protected communication between the desktop application and cloud infrastructure.

The authentication module uses OTP two-factor verification, which ensures that only authorized users can access the system. JWT token-based session management was implemented to maintain secure sessions for the entire period without having to validate the user's credentials each time. BCrypt hashing was used to create hashes of user passwords, thereby preventing exposure of user passwords in plain-text format.

Moreover, AWS IAM policies were employed to limit unauthorized access to AWS resources such as S3 storage and Rekognition services. Communication over cloud networks was performed using HTTPS/TLS protocols to ensure the integrity and confidentiality of sketches and recognition information transmitted between the application and the cloud server. These security mechanisms have enhanced the reliability of access control and reduced the likelihood of unauthorized individuals accessing sensitive forensic information.

E. System Performance Metrics

To evaluate the performance of the proposed solution, key focus areas included processing speed/recognition efficiency and operational reliability. To better understand execution performance characteristics and system responsiveness, each sub-process taking place during the workflow was analyzed.

The drag and drop sketch creation module enabled much quicker creation of suspect sketches than through more traditional hand-drawn forensics methods. Image preprocessing (cleaning and resizing) of the sketches was efficiently completed in OpenCV, resulting in much improved quality of by the time the sketch was uploaded for recognition.

The response times of AWS Rekognition for facial match operations yielded a quick response after sketch upload, and all recognition results were generated within a very short duration

after snapshot upload. Using the recognition engine at AWS Rekognition, high similarity scores and confidence ranges were produced in the case of properly generated sketches containing well-defined (clear) facial features.

The integrated workflow eliminated manual touch points (or at least reduced them) in the overall operational workflow, which improved operational efficiency by automating processes with respect to sketch processing, uploading to the cloud, extracting features from uploaded sketches, and matching suspects.

F. Scalability and Reliability Evaluation

Compared to traditional forensic sketching methods, the proposed cloud-integrated architecture is able to provide enhanced scalability and dependability in terms of the forensic investigation architecture. In addition to this, because the system is built upon the AWS cloud infrastructure, it has the capability of processing larger databases of criminal data and larger numbers of sketches without losing significant performance.

In particular, because of AWS S3, scalable storage management and efficient facial recognition (via AWS Rekognition) are available to handle large datasets of forensic sketches, and the modularity of the system will allow for greater ease of maintenance and future integration of advanced AI models, as well as potentially additional functionality for forensic investigations.

Additionally, the proposed system has demonstrated stable operational characteristics during repeated tests and provided consistent facial recognition performance from multiple sketch inputs. Cloud service integration has reduced dependence on local resources and improved the overall reliability of forensic investigations.

G. Summary of System Performance

From the evaluation of the proposed AI-Based Forensic Sketch Recognition System, it can be concluded that there is a considerable increase in the effectiveness of forensic suspect identification due to the use of image processing, cloud computing, and AI technology. The preprocessing using OpenCV increased the quality of sketches and ensured high similarity scores for face recognition.

The drag-and-drop feature of constructing a forensic sketch made it easier for the investigators without requiring the assistance of professional forensic artists. The use of cloud computing for storage and recognition services increased the scalability, efficiency, and accessibility of the forensic data. Security of the system was ensured through the use of OTP verification, JWT management, BCrypt Hashing, and IAM access control.

Therefore, it can be said that the AI-Based Forensic Sketch Recognition System modernized the traditional process of forensic sketch identification with a more secure and scalable AI-driven forensic investigation system.

VI. RESULTS AND DISCUSSION

The AI-based forensic sketch recognition system has been designed, developed, and tested for generating digital secure forensic sketches; to perform cloud-based facial recognition; and to identify possible suspects from eyewitness sketches. It accomplishes this goal by using various authentication methods, image preprocessing techniques, the Amazon Web Services (AWS) cloud platform, and AI recognition algorithms as part of a comprehensive forensic investigation platform.

The results of this study show that the system is efficient at generating composite facial sketches, preprocessing sketch images, securely uploading them to cloud storage, and accurately matching suspects from criminal databases using AWS Rekognition. The addition of OpenCV preprocessing techniques with cloud-based recognition greatly increases recognition compatibility among criminal photographs and forensic sketches.

A. Authentication and Security Results

The authentication module successfully validated users by validating user name/password combinations plus providing two factor authentication using an OTP delivered via email. One-time passwords were delivered to the user's authorized email accounts quickly after being issued, allowing users to have secure access to the forensic platform while conducting investigations.

Implementation of JWT session management and hashing passwords using bcrypt improved the protection of credentials and reduced the risk of the exposure of criminal investigation data. Identifying the systems through a variety of means (for example, Mac addresses and IP addresses) allowed for monitoring and verification of devices used to access the forensic platform.

The authentication process demonstrated an overall stable and secure performance throughout multiple rounds of testing.

B. Composite Sketch Generation Results

With the help from this drag-and-drop sketch development tool, investigators can now create forensic face sketches without having to have professional level of skill in drawing forensic faces. All users are provided with abilities to build their forensic faces through image components such as the head shape, eyes, nose, lips, eyebrows and hair based on what those who witness a crime describe about the person.

The modular approach to develop sketches has also decreased the complexity of getting to create a sketch and allows for increased consistency of the representation of a face shape/structure. Additionally, the functionally interactive interface allowed users to easily resize, move and change the facial components which give the ability to develop a sketch in a much shorter timeframe than conventional hand-drawn sketches.

Facial sketches developed for use in the program are very similar to the descriptions supplied by persons who have witnessed crimes, and sketched facial representations can

be used in conjunction with the Preprocessing Module and Recognition Module, which are both included in the system.

C. Image Preprocessing Results

The OpenCV-based Preprocessing module helped to enhance the quality of the Forensic Sketches produced prior to any recognition process. Grayscale Conversion of the sketches resulted in a less complex computational workload and also eliminated distracting colored pixels from the sketch images.

Contrast enhancement using CLAHE helped to improve facial features (eyes, nose, mouth, jawline) in the sketches, while the Noise Reduction Techniques helped to reduce the distortion of the sketches and improve their overall smoothness. Experimental analysis indicated that the use of Preprocessing Techniques improved Overall Clarity of Facial Features within Forensic Sketches, and also increased the likelihood that Forensic Sketches would receive a higher Probability Match to AWS Rekognition Recognition.

Based on the comparison of results from Preprocessed Sketches and Raw Forensic Sketches, Preprocessed Sketches produced higher similarity scores and superior success rates.

D. Cloud Storage and Recognition Results

The proposed system utilized AWS S3, AWS Rekognition, and cloud technologies to successfully identify alleged suspects. Forensic sketch data was securely transmitted and stored in AWS S3 using encrypted transport protocols and cloud access security tools.

The processing of uploaded forensic sketches was accomplished through the use of AWS Rekognition. This was done by extracting facial embeddings from the forensic sketch data and performing a comparison between each sketch and the criminal images in the server's database of images. A successful match for each sketch retrieved the criminal record correspondingly with a similarity percentage and a confidence score.

The performance of the system was measured by the efficiency of communication in the cloud and the ability to recognize multiple sketches in multiple trials in repeated processes. The cloud-based architecture allowed for scalability and reduced reliance on locally-based computational capability.

E. Face Matching and Recognition Analysis

The face recognition module provided many successful test cases between the uploaded forensic sketch and the stored criminal mugshot, producing a high level of confidence for similarity matches between sketches that were correctly constructed and pre-processed.

The recognition results provided include:

- Mugshot with a documented record.
- Record showing the mugshot's record number.
- Similarity percentage between the sketch and the mugshot.
- Confidence score generated by the recognition system.

The sketches with good facial structure and better pre-processing produced the highest degree of similarity and recognition accuracy. The AWS Rekognition has produced efficient matching performance when comparing the sketch to the actual facial photograph, even if the two looked different.

The recognition process saved time for the manual investigation and significantly decreased the time taken to identify a suspect from the initial recognition through to their identification for a well-structured forensic investigation.

F. System Performance Analysis

The forensic system being proposed exhibited excellent operational efficiency across all modular functionality. The drag-and-drop capability for sketch creation created efficiencies in the time required to create sketches with non-technical investigators. Additionally, the image enhancement processes performed in the preprocessing module were able to operate efficiently without causing any significant processing delays.

Moreover, the use of AWS cloud services provided reliable communication with high availability, scalable storage, and improved response time for facial recognition.

In addition, the automated workflow of the system provided the following advantages:

- Value-added security through authentication mechanisms.
- Sketch construction using facial feature selection.
- Image preprocessing for quality enhancement.
- Secure image uploading to cloud storage.
- Facial recognition and suspect matching.
- Result retrieval with similarity and confidence scores.

Finally, the modular architecture of the proposed forensic system will improve overall maintainability, scalability, and operational reliability. Additionally, the proposed system will eliminate the traditional relationship of dependency between forensic sketching and the underlying methods used to create forensic sketches.

G. Overall Analysis

Evaluation of the proposed AI-Based Forensic Sketch Recognition System supports its functional capability as a reliable, secure and scalable solution to today's modern forensic investigation requirements. By using an effective combination of image processing, cloud computing and artificial intelligence to identify suspects in forensic open cases, the effectiveness of the proposed system for suspect identification was significantly improved over that of traditional manual methods.

The drag-and-drop mechanism used to build composite sketches made it easier to generate facial composite sketches, and the image preprocessing functionality offered by OpenCV improved the recognition compatibility of composite sketches with criminal images. The reliability of matching composite sketches to the corresponding criminal images was demonstrated by the use of AWS Rekognition facial recognition service which produced facial matching results with high similarity confidence scores, and the use of AWS S3 allowed for the secure centralized storage of forensic records.

The proposed system has successfully modernized the traditional methods of identification of suspects in forensic investigations through reduced time to identify suspects, improved accuracy of the identification of suspects and support for secure cloud storage of criminal records.

VII. CONCLUSION AND FUTURE SCOPE

A. Conclusion

The Proposed AI-based Forensic Sketch Recognition System combines multiple components of the forensic identification process (sketch generation, image pre-processing, cloud computing and artificial intelligence) into one cohesive system for the purpose of identifying criminal offenders. The new system was designed to address the shortcomings of traditional forensic sketching methods which are often tedious and depend heavily on task completion by forensic artists. Additionally, traditional sketching methodologies make integration with modern-day methods of digital recognition very difficult.

Investigators will be able to create composite facial sketches using the proposed system's 'drag and drop' interface, without the requirement of having advanced drawing skill. Sketches generated by the system will be enhanced by image pre-processing techniques utilizing OpenCV, including: conversion to grayscale, enhancement of contrast through CLAHE algorithms, and noise reduction to improve the quality of the completed images and increase the likelihood that the images can be identified within existing databases.

The successful integration of AWS S3 cloud storage and AWS Rekognition provided the system with scalable and efficient methods to compare "sketch to photo". The proposed system was able to identify suspect(s) by comparing the forensic sketches generated using the system with the existing databases of criminal mug shots, generating a similarity score and providing the level of confidence for each comparison. The proposed system incorporated multiple security mechanisms to include: OTP-based authentication; JWT session management; BCrypt hash password storage; and access control via the cloud storage provider in order to ensure the appropriate handling of all sensitive forensic information.

The experiment conducted to evaluate how well the suggested solution helps to decrease the length of time it takes to identify suspects gives rise to many benefits, such as improving investigations' efficiency and reducing manual labor involved in forensic operations. With its cloud-based architecture, modular design, and AI-enabled cognitive capabilities, the system is an ideal fit for contemporary law enforcement and forensic investigation; thus as a whole, the provided solution successfully modernizes traditional forensic sketch identification techniques by providing an intelligent, secure, and scalable platform on which to conduct all related activities.

B. Future Scope

While the proposed system shows impressive capabilities in recognizing forensic sketches, there are still several future

enhancements that can occur that will increase the effectiveness and scalability of the system as well as provide more applicable results in real-world scenarios.

Future enhancements can consist of incorporating advanced deep learning models like CNN's and GAN's that will increase the accuracy of transforming sketches to photos and recognizing that photo. The addition of real-time face recognition or surveillance functionality could be useful in allowing for real-time suspect tracking and ultimately more accurate automatic criminal detection.

Since the current system is primarily designed for a desktop environment, future development will extend it to create web-based and mobile forensic platforms that are more accessible and provide a larger resource for remote-based investigations. Furthermore, future developments may include the addition of other forms of biometric verification such as finger and iris recognition to improve the multi-faceted identification of criminals.

Possibilities for enhancements include:

- Automatic generation of sketches from verbal descriptions.
- AI-based systems for determining suggested facial characteristics.
- Large database of criminal offenders.
- Real-time cloud synchronization across police departments.
- Advanced forensic reporting and analysis.
- Multi-language support for wider usability.
- Integration with national crime record management systems.

Further improvements to preprocessing and recognition algorithms will improve recognition capabilities of sketched faces created from low-quality sketches and various orientations. More sophisticated artificial intelligence techniques and expanded cloud infrastructures can change the proposed system into a fully intelligent criminal identification and forensic investigation tool for future law enforcement use.

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